What does the logit value actually mean?

Asked 11 years, 3 months ago Modified 3 years, 7 months ago Viewed 63k times

47

I have a logit model that comes up with a number between 0 and 1 for many cases, but how can we interprete this?

Lets take a case with a logit of 0.20

Can we assert that there is 20% probability that a case belongs to group B vs group A?

is that the correct way of interpreting the logit value?

regression logistic interpretation

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edited Nov 27, 2020 at 12:10
kjetil b halvorsen ♦
81.3k 32 199 646

asked Mar 20, 2013 at 15:32
Dez
471 1 5 3

4 In addition to @SvenHohenstein's good answer below, it may help you to read my answer here: [Interpretation of simple predictions to odds ratios in logistic regression](), which contains additional basic information about probabilities & odds. Note that the logit can be understood more abstractly as a link function; you can read more about that here: [difference-between-logit-and-probit-models]() (although this answer might be a bit more technical). – gung - Reinstate Monica Mar 20, 2013 at 23:08

1 I want to know why the pronunciation of *logit* is neither like *logarithm* nor like *logistic* – Henry Nov 15, 2013 at 1:14

@Henry - According to Wiktionary, the U.S. pronunciation of 'logit' /ˈloʊdʒɪt/ (en.wiktionary.org/wiki/logit) is like 'logistic' (/loʊˈdʒɪs.tɪk/) (en.wiktionary.org/wiki/logistic). – shaneb Feb 8, 2019 at 22:39

1 @shaneb - fair enough - thought that only shifts the question to the un*logical* pronounciation of *logistic* – Henry Feb 10, 2019 at 20:38

3 Answers

Sorted by: Highest score (default)

81

The logit L of a probability p is defined as

$$L = \ln \frac{p}{1-p}$$

The term $\frac{p}{1-p}$ is called odds. The natural logarithm of the odds is known as log-odds or *logit*.

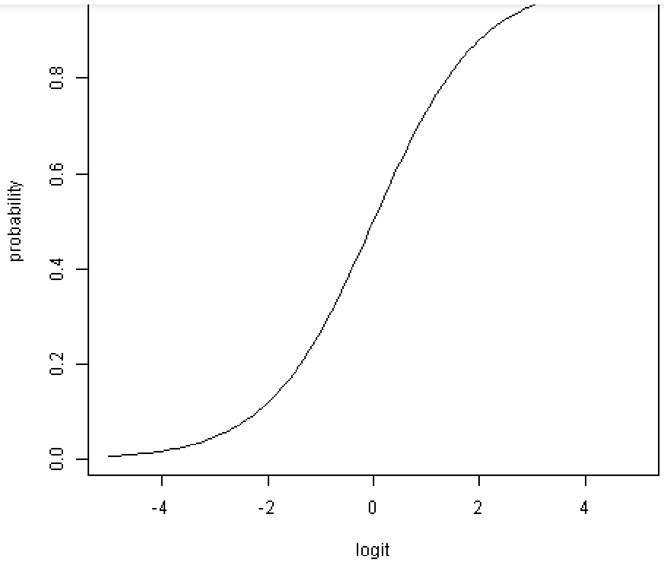
The inverse function is

$$p = \frac{1}{1+e^{-L}}$$

Probabilities range from zero to one, i.e., $p \in [0,1]$, whereas logits can be any real number ($\mathbb{R}$, from minus infinity to infinity; $L \in (-\infty, \infty)$).

A probability of 0.5 corresponds to a logit of 0. Negative logit values indicate probabilities smaller than 0.5, positive logits indicate probabilities greater than 0.5. The relationship is symmetrical: Logits of -0.2 and 0.2 correspond to probabilities of 0.45 and 0.55, respectively. Note: The absolute distance to 0.5 is identical for both probabilities.

This graph shows the non-linear relationship between logits and probabilities:



The answer to your question is: There is a probability of about 0.55 that a case belongs to group B.

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edited Nov 15, 2013 at 1:04
Nick Cox
58.5k 8 132 198

answered Mar 20, 2013 at 16:51
Sven Hohenstein
6,913 26 35 42

" *Logits of* -0.2 *and* 0.2 *correspond to probabilities of* 0.45 *and* 0.55 *respectively.*" How does it imply that logit distribution is symmetric? – user 31466 Feb 12, 2015 at 3:50

There is a probability of about 0.55 *that a case belong to group B.* When will it belong to group A ? – user 31466 Feb 12, 2015 at 3:55

2 @Leaf Since there are only two groups, A and B, the probability for group A is $1 - 0.55 = 0.45$. – Sven Hohenstein Feb 12, 2015 at 6:59

1 @Here, symmetry is related to the *absolute difference* to a probability of 0.5 or a logit of 0. If probability is $0.5 + x$, the logit is $0 + y$; If probability is $0.5 - x$, the logit is $0 - y$. Here, $\text{sign}(x) = \text{sign}(y)$. – Sven Hohenstein Feb 12, 2015 at 7:02

5

To add a more modern (but not very deep) perspective, consider how it's used in deep learning (ha, pun intended...):

logit is referred to the output of a function (e.g. a Neural Net) just before it's normalization (which we usually use the softmax). This is also known as the code. So if for label y we have score $f_y(x)$ then the logit is:

$$logit = \log\left(\frac{e^{f_y(x)}}{Z}\right) = score = f_y(x)$$

Where Z is the standard partition function. By the way, this is all over the place in the pytorch and tensorflow documentation.

So you can interpret it as:

> **the (unnormalized) score for a label or (functional confidence) for a specific class/label.**

One of the many references: https://stackoverflow.com/questions/41455101/what-is-the-meaning-of-the-word-logits-in-tensorflow

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answered Dec 1, 2020 at 21:03
Charlie Parker
6,916 14 75 130

http://www.ats.ucla.edu/stat/stata/seminars/stata_logistic/default.htm

http://www.ats.ucla.edu/stat/stata/dae/logit.htm

http://www.ats.ucla.edu/stat/stata/faq/oratio.htm

http://www.ats.ucla.edu/stat/mult_pkg/faq/general/odds_ratio.htm

so if the coefficient is 0.2 it depends on the variable, I guess you have a dummy, which is e.g. 0 for group B and 1 for group A?

odds ratio is given by: $OR = e^b$

so in your case: $e^{70.20}$

This would be the odds ratio of your group variable corresponding to your reference group.

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answered Mar 20, 2013 at 15:59
Stat Tistician
2,341 5 34 57

3 I believe the OP is asking about how to *interpret* logits, not how to perform logistic regression. – whuber ♦ Mar 20, 2013 at 16:22